



School of Computer Science
COMP 2067 Programming for Beginners
Summer 2025 – Assignment 2
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For each question, write a program with comments to perform the required tasks.

Question 1.

- a. Initialize a string *str1*. This string *str1* should contain the name of a well-known Canadian city or landmark, with 6-12 characters. Print *str1*.
- b. Ask the user to enter a positive integer between **1** and **13** and assign it to a variable *m*. Print *m*. Assume the user will enter an integer, but it may not be in the proper range.
- c. Check if *m* is in the proper range. If *m* is in the proper range:
 - i. Create a variable *x* and assign it the value of the *m*th province/territory's first letter, based on alphabetical order (['A', 'B', 'M', 'N', 'N', 'N', 'N', 'N', 'O', 'P', 'Q', 'S', 'Y']). Convert *x* to lowercase. Print *x* with a *suitable message*¹.
 - ii. Find the number of occurrences of *x* in *str1* and print this value with a *suitable message*¹. Otherwise, print a message saying that the user entered an invalid input.
- d. Print *str1* centered in a field of 25 characters, with ~ used as padding.

Question 2.

- a. Ask the user to enter a positive integer *n* between 1867 and 2025 (representing a year in Canadian history). You may assume the user will enter an integer, but it might not be in the proper range. Print *n*.
- b. Check the value entered to see if *n* is in the proper range.
- c. If the user entered a year between 1867 and 2025, do the following:
 - i. If *n* is a year in which Canada hosted the Olympics (1976, 1988, or 2010), print *n* with a message indicating Canada hosted the Olympics in that year.
 - ii. If *n* is a year of a major Canadian milestone (e.g., 1867 - Confederation, 1965 - New Flag, 1982 - Constitution Act), print *n* with a message indicating the historical event.
 - iii. If *n* is divisible by 4 but not in the above categories, print *n* with a message indicating it was a leap year in Canada.
 - iv. Otherwise, print *n* with a message indicating that nothing special is recorded for that year.
- d. Otherwise (i.e., *n* is not in the proper range), print the message: "You did not enter a valid input!"

Question 3.

- a. Define a list **L1** consisting of *at least 11 famous Canadian cities or landmarks* (e.g., L1 = ["Windsor", "Toronto", "Vancouver", "Banff", "Quebec City", "Niagara Falls", "Montreal", "Ottawa", "Whistler", "Halifax", "Jasper"]) and print **L1**.
- b. Ask the user to enter an integer **num** between **0** and **max-1**, where **max** is the number of elements in the list. You may assume the user will enter a valid input.
- c. Remove the element with index **num** from **L1** and print the **removed element**. Also, print the **updated L1** (after removing the element with index **num**).
- d. If **num** is greater than or equal to 5:
 - i. Print the **first num elements** of the updated **L1** with a *suitable message*¹.
- e. Otherwise, print the last **num** elements of the updated **L1** with a *suitable message*¹.

Submission:

1. You will earn a maximum of 100 points (accounts for 7%) for submitting your **source code** (*lastname_sol_A2.py*) within the specified deadline.
2. **Do not upload your source code to Brightspace as a single zip/rar file**. Such submissions will not be accepted and a mark of **zero** will be assigned.
3. Marks will be **deducted** if *comments and/or a suitable message*¹ are missing.
4. **Late submissions** will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.
5. **Unlimited resubmissions** are allowed. However, only the last submission will be considered. Note that if you resubmit after the deadline, a penalty will apply, even if an earlier version was submitted on time.

¹**NOTE:** In this course, the phrase '*suitable message*' refers to some kind of description that describes what is being printed. For example, just printing the value **22** by itself does not tell the user anything useful; however, printing '**x = 22**' would add some context to what is being printed and so this would qualify as a '*suitable message*' if the value of x is to be printed.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.